



AI PREDICTION MARKET INTELLIGENCE AGENT

TURNING PREDICTION-MARKET DATA INTO
STRUCTURED, EVIDENCE-BASED ANALYSIS

BUILT WITH NBN • POLYMARKET API
• AI AGENT

PROJECT OVERVIEW

What I Built & Why

The Project

I built an AI Prediction Market Intelligence Agent that takes a natural-language prediction-market question, retrieves relevant market data from Polymarket, evaluates the market context, and produces a structured analysis.

The Problem

Prediction-market questions can look simple, but finding the right market and interpreting its information correctly requires more than keyword matching.

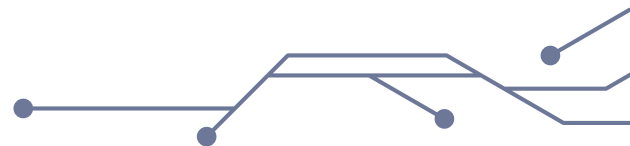
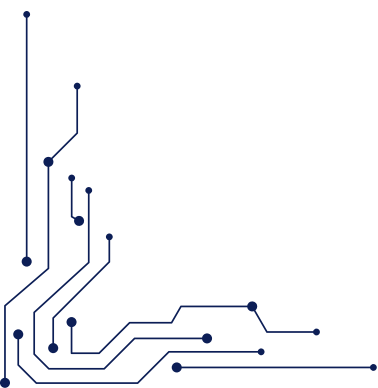
A market may appear relevant while having different:

- Thresholds
- Dates
- Resolution conditions
- Market status
- Outcome definitions

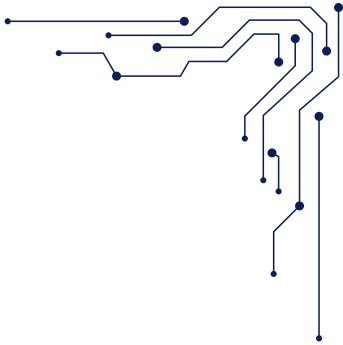
The Approach

I designed the Agent to:

- Understand the question
- Preserve the user's original question and key conditions.
- Find relevant market data
- Search Polymarket for candidate markets.
- Validate the match
- Consider relevance, market status, resolution criteria, end date, and outcome information.
- Interpret the data
- Use market prices, volume, and liquidity as contextual signals.
- Produce structured analysis
- Separate market-implied probability, evidence, uncertainty, and conclusions.



How It Works



Component	Purpose
User Question	Receives the prediction-market question
Edit Fields	Standardises the question
Polymarket API	Retrieves relevant market data
AI Agent	Evaluates the market and evidence
Structured Output	Organises the analysis
Market Analysis	Presents the final result

```
18 {
19   {
20     "id": "d2e5a8f9-0fa0-4841-8495-a4b652927f81",
21     "name": "question",
22     "value": "Will Bitcoin reach $150000 at the end of 2026?",
23     "type": "string"
24   },
25   {
26     "id": "48ae6e8b-01a1-494a-94f4-a00520107e9f",
27     "name": "question ",
28     "value": "={{ $json.chatInput }}",
29     "type": "string"
30   }
31 },
32 "options": {}
33 },
34 "type": "n8n-nodes-base.set",
35 "typeVersion": 3.5,
36 "position": [
37   208,
38   -48
39 ],
40 "id": "cd7a0a62-1a5b-424a-8ce2-e3931c889d45",
41 "name": "Edit Fields"
42 },
43 {
44   "parameters": {
45     "promptType": "define",
46     "text": "=User question:\n{{ $('Edit Fields').item.json.question }}\n\nPolymar
47     "hasOutputParser": true,
48     "options": {
49       "systemMessage": "CRITICAL QUESTION INTEGRITY RULE:\n\nPreserve the user's q
```

Technology: n8n • Polymarket API • AI Agent • Structured Output



Market Intelligence & Decision Logic

The Agent does not treat the first keyword match as the correct market.
It evaluates the available market information before using it in the analysis.

1. Question Relevance

Does the market describe the same event and threshold?

2. Market Status

Is the market active/open and suitable for current analysis?

3. Resolution Criteria

What conditions determine whether the market resolves YES or NO?

4. End Date

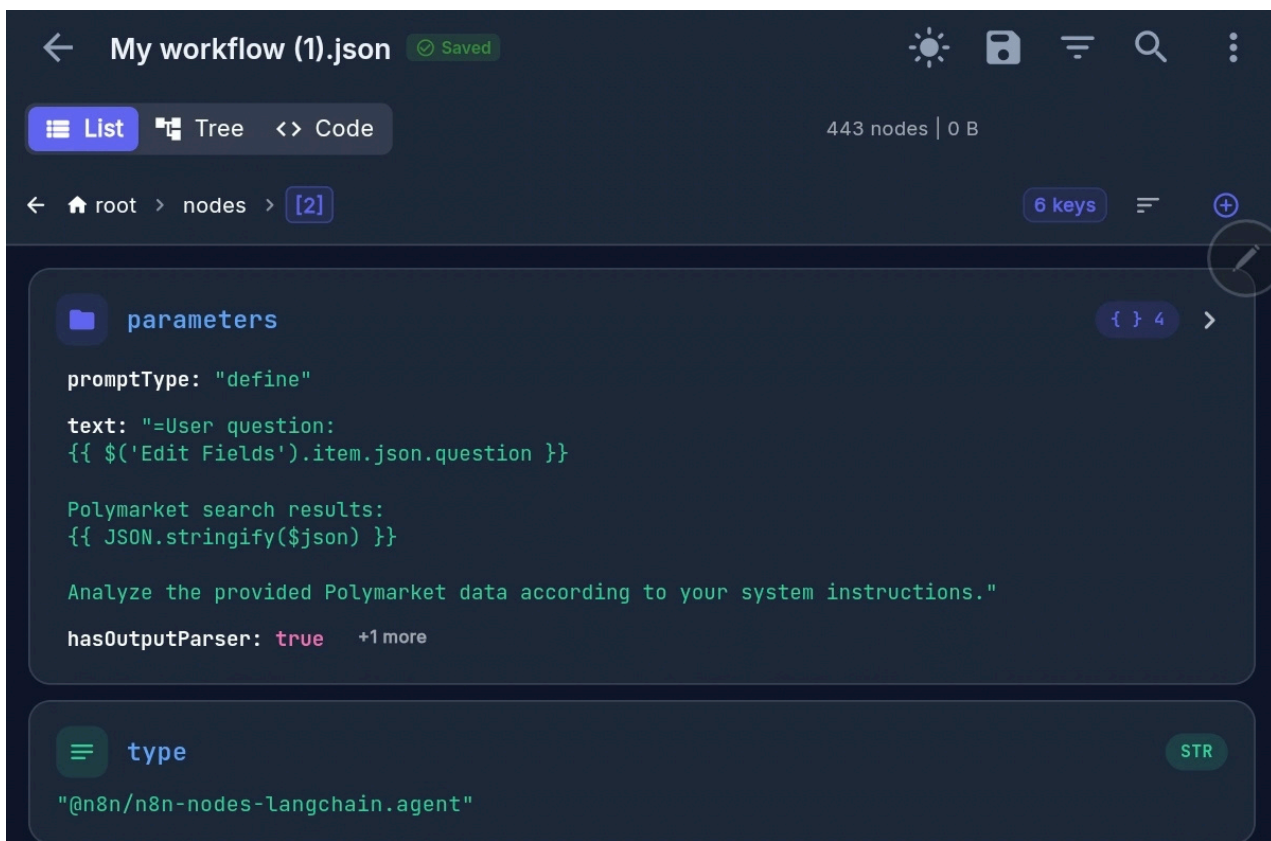
Does the market's time period match the user's question?

5. Outcome Prices

What do the YES/NO prices indicate about the market-implied probability?

6. Volume & Liquidity

What additional context do trading activity and available liquidity provide?



AI Agent Configuration

Real Test & Results

Test Question

Will Bitcoin reach \$150,000 at the end of 2026?

Market Identified

Will Bitcoin reach \$150,000 by December 31, 2026?

The Agent identified a directly corresponding Polymarket market.

Observed Market Data

Metric	Result
YES Price	\$0.0285
Market-Implied Probability	≈ 2.85%
Trading Volume	> \$1.18M
Liquidity	≈ \$100K
Market Deadline	December 31, 2026

What the Test Demonstrated

The workflow successfully:

- Received the natural-language question
- Identified a corresponding market
- Retrieved real market information
- Interpreted the YES price as a market-implied probability
- Produced structured analysis

Validation Note

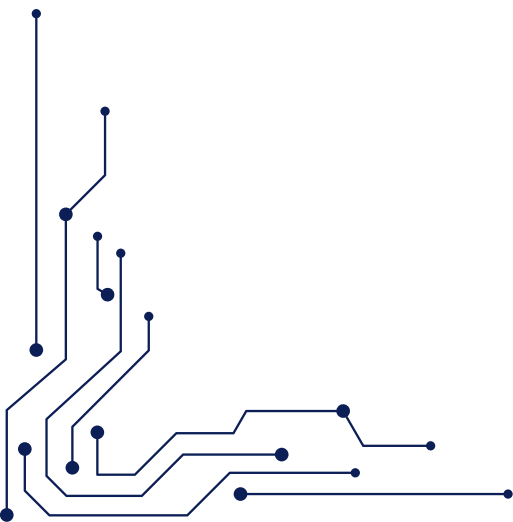
The observed 2.85% represents the probability implied by the market price at the time of the test.

It is a market signal, not as a guaranteed or objectively correct probability.

2.85% | \$1.18M+ | ~\$100K | Dec 31, 2026

Development Finding

In this test, the Agent's estimated probability matched the market-implied probability. This highlighted an area for future improvement: developing a more clearly independent probability-estimation process.



SKILLS & PROJECT VALUE

What This Project Demonstrates

AI & Automation

- AI Agent workflow design
- n8n workflow automation
- Structured AI outputs

APIs & Data

- External API integration
- HTTP requests
- JSON data handling
- Prediction-market data interpretation

Research & Analysis

- Market research
- Semantic market matching
- Probability interpretation
- Evidence-based reasoning
- Uncertainty analysis

What I Built?

An end-to-end AI-powered data workflow

Question → Data → Evaluation → Reasoning → Structured Analysis

The project demonstrates the ability to research an unfamiliar domain, integrate external data, build an automated workflow, test it with real data, identify limitations, and document the development process.

FUTURE DEVELOPMENT

Independent probability estimation

Additional evidence sources

Stronger market matching

Historical market analysis

Automated monitoring

PROJECT STATUS

Working First Version

Built • Tested • Documented

Designed for continued development.

